

The Story of Menopur (Menotropin) on Earth

A cited chronological history & evolution of highly purified hMG (HP-hMG)

INN: menotropin · Brand: Menopur (Ferring) · Compiled 2026-06-27 · Medical Affairs reference

Evidence note. Built from primary sources (Human Reproduction, Fertility & Sterility, Cochrane, Hum Reprod Open, PMC). 3 claims passed full adversarial verification; others are direct quoted extractions from primary journals but the automated cross-check was cut short by a token limit ([sourced]). Unconfirmed items are flagged [GAP]. Verify against source before external use.

The Narrative

The problem hMG solved. For most of history there was no way to stimulate the ovaries. The breakthrough came from the urine of postmenopausal women, rich in FSH and LH. hMG (INN menotropin) was first extracted around 1949–50; the first preparation (Pergonal, Serono) followed. [GAP: exact year varies by source]

1961 — the door opens. Bruno Lunenfeld's group delivered the first child after hMG ovarian stimulation in 1961, the event that opened the path to controlled ovarian stimulation — the foundation of modern IVF. [VERIFIED]

Late-1980s safety reckoning. Human *pituitary* gonadotropin (HPG) was withdrawn in the late 1980s after a link to Creutzfeldt–Jakob disease, leaving urinary hMG dominant — but early hMG carried many unknown urinary proteins, making quality control almost impossible. [VERIFIED]

Purification → Menopur. Ferring developed highly purified hMG (HP-hMG = Menopur), standardised to 75 IU FSH + 75 IU LH activity per vial, the LH activity supplied mainly by hCG. This 1:1 FSH:LH-activity design is Menopur's signature vs pure recombinant FSH. [sourced]

The evidence era (2006–2020). Tested head-to-head vs recombinant FSH in MERiT (2006), Coomarasamy (2008), Cochrane (2011), MEGASET (2012) and MEGASET-HR (2020): Menopur is at least as effective for live birth, with signals of better embryo quality and, in high responders, less OHSS and pregnancy loss.

The convenience era (2018–21+). The latest chapter is delivery: Menopur moved from powder-for-reconstitution to a stable liquid pre-filled pen (31G needle), shown bioequivalent to the powder. [VERIFIED]

Where it stands today. Cochrane evidence and guidelines treat rFSH and HP-hMG as broadly equivalent for live birth; choice is driven by patient profile, OHSS risk, convenience and cost — Menopur's edge being its LH activity and OHSS profile in high responders.

Year-by-Year Timeline

Year	Milestone	Source / status
~1949–50	hMG (menotropin) first extracted from postmenopausal urine; Pergonal/Serono lineage	PMC6616070 · [GAP] year
1961	First child after hMG stimulation (Lunenfeld) — birth of controlled ovarian stimulation	VERIFIED · Lunenfeld 2012
Late 1980s	Pituitary gonadotropin (HPG) withdrawn after CJD link; urinary hMG dominant	VERIFIED · Lunenfeld 2012
1990s–2000s	Purification → HP-hMG / Menopur; 75 IU FSH + 75 IU LH activity (LH from hCG)	sourced · PMC8594316; FDA
2006	MERiT RCT (Andersen) — HP-hMG vs rFSH, long agonist	sourced · Hum Reprod
2007	Ziebe embryo-quality analysis of MERiT	sourced · Hum Reprod
2008	Coomarasamy meta-analysis — hMG live-birth edge	sourced · Hum Reprod/RBMO
2011	Cochrane (van Wely) — rFSH vs urinary, no live-birth difference	sourced · CD005354.pub2
2012	MEGASET RCT (Devroey) — antagonist, single blastocyst	sourced · Fertil Steril
2019	Bordewijk meta-analysis — near-equivalent dosing	sourced · Hum Reprod Open
2020	MEGASET-HR RCT (Witz) — high responders; less OHSS	sourced · Fertil Steril
2018–21	Liquid pen formulation; bioequivalent to powder	VERIFIED · PMC8594316
2025–26	rFSH ≈ HP-hMG for live birth; choice by profile/OHSS/cost	Cochrane + guidelines

[GAP] not confirmed this run: exact US FDA approval date, EU approval dates, and label/indication change history of Menopur. (A Menopur+Bravelle mixed-protocol FDA approval appears only in a low-reliability press release — verify on FDA/DailyMed.)

The Pivotal Trials

MERiT — 2006 · Andersen et al., Hum Reprod 21(12):3217

Randomized, assessor-blind, multinational RCT. **731** IVF women (HP-hMG 363 / rFSH 368), long GnRH agonist. Primary endpoint ongoing pregnancy: **27% vs 22%**, OR 1.25 (95% CI 0.89–1.75) — non-inferiority established, superiority not concluded. [sourced, not cross-verified]

Ziebe (MERiT embryo quality) — 2007 · Hum Reprod 22(9):2404

Top-quality embryos/oocyte higher with HP-hMG: **11.3% vs 9.0% (P=0.044)** local, NS central. Among top-quality embryos, live birth/ongoing pregnancy **48% vs 32% (P=0.038)** — signal of better implantation capacity. [sourced, not cross-verified]

Coomarasamy meta-analysis — 2008 · Hum Reprod 23(2):310

7 RCTs, **2159 women**, hMG vs rFSH, long agonist. Live birth RR **1.18 (95% CI 1.02–1.38, P=0.03)** — ~4% absolute edge for hMG. [sourced, not cross-verified]

Cochrane review — 2011 · van Wely, CD005354.pub2

42 RCTs, 9606 couples. No significant live-birth difference: **OR 0.97 (95% CI 0.87–1.08)**. Choose by availability, convenience, cost. [sourced, not cross-verified]

MEGASET — 2012 · Devroey et al., Fertil Steril

Open-label, assessor-blind, **749 women**, 25 centres/7 countries, GnRH antagonist, single-blastocyst transfer. Ongoing pregnancy **30% vs 27%** (non-inferior). Cumulative live birth (incl. frozen ≤1 yr) **40% vs 38%**. [sourced, not cross-verified]

MEGASET-HR — 2020 · Witz et al., Fertil Steril

620 predicted high responders (AMH ≥5 ng/mL), antagonist ART/ICSI. Ongoing pregnancy (fresh) **35.5% vs 30.7%** (non-inferior). **OHSS 9.7% vs 21.4%**; early pregnancy loss 14.5% vs 25.5%. Cumulative live birth similar (50.6% vs 51.5%). [sourced, not cross-verified]

Bordewijk meta-analysis — 2019 · Hum Reprod Open hoz008

28 RCTs, **7553 women**. Total gonadotrophin dose differed only marginally (MD –37 IU) — near-equivalent dosing. [sourced, not cross-verified]

Composition & Mechanism

Menotropin (Menopur, HP-hMG) = highly purified urinary preparation, **75 IU FSH + 75 IU LH activity per vial (1:1)**; the LH activity is supplied mainly by hCG (~10 IU hCG per 75 IU FSH) plus low native LH. FSH drives follicular recruitment; hCG-driven LH activity supports theca-cell androgen production and steroidogenesis — the basis for interest in LH activity in selected patients. [sourced; exact hCG:FSH ratio not cross-verified — GAP]

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